



Sex and the single snail

An exploration of the possible advantages of sexual reproduction over asexual reproduction. <https://digit.in/nov19-03>



The Amazon Molly

Find out more about the Amazon Molly and how the fish has successfully adopted asexual reproduction. <https://digit.in/nov19-04>

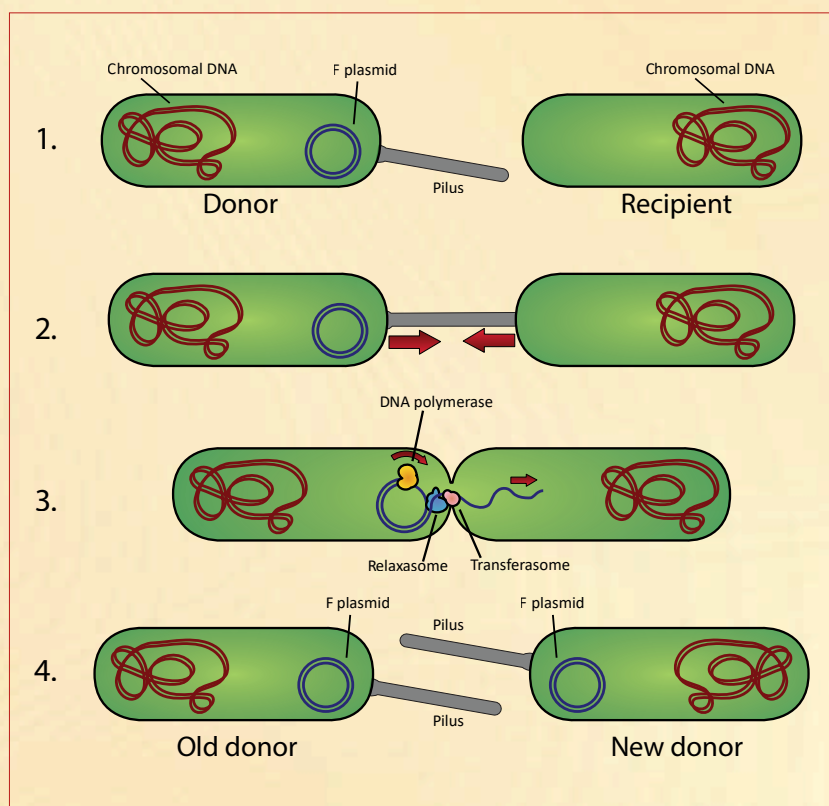
HORIZONTAL GENE TRANSFER

The simplest living organisms, such as bacteria, exhibit behaviours that are classified as parasexuality. This is an umbrella term for mechanisms by which cells transfer genetic data to other cells, without sexual reproduction. The genetic material is transferred from one creature to another, without the creation of a new generation. *E. coli* bacteria exhibit what is known as bacterial conjugation. Here, a donor cell produces a structure that functions as a genetic bridge, known as the pilus. The pilus attaches to a recipient cell, and then brings the two cells together. Enzymes that replicate DNA kick in, and genetic material is transferred from the donor to the recipient. Then, the recipient grows a pilus before separation, and can now also function as a donor cell. Another process of horizontal gene transfer is known as transformation. Here, bits and pieces of DNA are absorbed from the surrounding environment, directly through the cell walls. The DNA strands are then processed by the cell organelles, and incorporated into the DNA. Bacteria can pick up the genetic information of dead cells in their vicinity, through this technique. The third process is known as transduction. Here, the gene transfer occurs through the agency of a virus, in the case of bacteria, known as bacteriophages. The bacteriophages infect bacteria to utilise their genetic replication machinery. During this infection, genetic material from the bacteria and the virus intermingle, and strands from one may be left within the other. In some rare cases, the bacterial DNA may get packaged within the virus. When the phage subsequently infects another cell, the genetic material may be transferred. Within the new host, the material can be cut up and recycled according to the designs of the host DNA, or, in case a suitable portion of the DNA is found where the new strand can fit in perfectly, it does. All these mechanisms of horizontal gene transfer are called parasexualities, and are considered to be the precursor to the very origin of sexual reproduction.

THE THEORIES

In Lewis Carroll's *Through the Looking Glass*, the Red Queen eggs Alice to keep running as fast as she can, just to remain in the same place. In science, according to the Red Queen theory, a species has to evolve as rapidly as it can, to keep pace with another co-evolving species. For example, both the birds and the bees are evolving to be faster at flight, the bees to escape and the birds

changes in space. A species that moves from the sea shore to the mountains may be able to adapt more rapidly to the higher altitudes if it reproduces sexually, as against a species that reproduces asexually. Simulations have shown that organisms that are well adapted to their environment stand to benefit if they reproduce asexually, while organisms that are not that well adapted to the environment stand to benefit if they



Bacterial conjugation

to prey. Yet, over long periods of time, the relation between the two remains the same. The predator species can be considered as exerting an environmental pressure on the prey species, and this environment changes rapidly. In such cases, the favourable traits built up by previous generations may no longer be advantageous. For the species, it is better to have a pool of genetic material available for recombination, which improves the fitness of each generation. The Red Queen theory explains the emergence of sexual reproduction as an effect of changes over time, and a similar theory explains it as an effect of

reproduce sexually. The Amazon molly is an example of a hybrid fish that won the genetic lottery. Between 100,000 to 200,000 years ago, an Atlantic molly mated with the Sailfin molly to produce the Amazon molly, with an incredibly hardy genetic make up. The fish pretends to mate with a male of a related species, which triggers the process of reproduction, but only the genetic material of the mother is transferred to the offspring. For over 500,000 generations, the fish has been reproducing asexually, without any deterioration in the genetic code. However, the Amazon molly is very much the exception. ❏